3.2 WEIGHTED LEAST SQUARES ESTIMATION

In the previous section, we assumed that we had an equal amount of confidence in all of our measurements. Now suppose we have more confidence in some measurements than others. In this case, we need to generalize the results of the previous section to obtain weighted least squares estimation. For example, suppose we have several measurements of the resistance of an unmarked resistor. Some of the measurements were taken with an expensive multimeter with low noise, but other measurements were taken with a cheap multimeter by a tired student late at night. We have more confidence in the first set of measurements, so we should somehow place more emphasis on those measurements than on the others. However, even though the second set of measurements is less reliable, it seems that we could get at least *some* information from them. This section shows that we can indeed get some information from less reliable measurements. We should never throw away measurements, no matter how unreliable they may be.

To put the problem in mathematical terms, suppose x is a constant but unknown n-element vector, and y is a k-element noisy measurement vector. We assume that each element of y is a linear combination of the elements of x, with the addition of some measurement noise, and the variance of the measurement noise may be different for each element of y:

$$\begin{aligned} \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} &= \begin{bmatrix} H_{11} & \cdots & H_{1n} \\ \vdots & \ddots & \vdots \\ H_{k1} & \cdots & H_{kn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_k \end{bmatrix} \\ E(v_i^2) &= \sigma_i^2 \quad (i = 1, \ldots, k) \end{aligned} \tag{3.11}$$

We assume that the noise for each measurement is zero-mean and independent. The measurement covariance matrix is

$$\begin{aligned} R &= E(vv^T) \\ &= \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & \sigma_k^2 \end{bmatrix} \end{aligned} \tag{3.12}$$

Now we will minimize the following quantity with respect to $\hat{x}$.

$$J = \epsilon_{y1}^2/\sigma_1^2 + \cdots + \epsilon_{yk}^2/\sigma_k^2 \tag{3.13}$$

Note that instead of minimizing the sum of squares of the ϵ_y elements as we did in Equation (3.4), we will minimize the *weighted* sum of squares. If y_1 is a relatively noisy measurement, for example, then we do not care as much about minimizing the difference between y_1 and the first element of $H\hat{x}$ because we do not have much confidence in y_1 in the first place. The cost function J can be written as

$$\begin{aligned} J &= \epsilon_y^T R^{-1} \epsilon_y \\ &= (y - H\hat{x})^T R^{-1} (y - H\hat{x}) \\ &= y^T R^{-1} y - \hat{x}^T H^T R^{-1} y - y^T R^{-1} H\hat{x} + \hat{x}^T H^T R^{-1} H\hat{x} \end{aligned} \tag{3.14}$$

Now we take the partial derivative of J with respect to $\hat{x}$ and set it equal to zero to compute the best estimate $\hat{x}$:

$$\begin{aligned}
\frac{\partial J}{\partial \hat{x}} &= -y^T R^{-1} H + \hat{x}^T H^T R^{-1} H \\
&= 0 \\
H^T R^{-1} y &= H^T R^{-1} H \hat{x} \\
\hat{x} &= (H^T R^{-1} H)^{-1} H^T R^{-1} y
\end{aligned} \tag{3.15}$$

Note that this method requires that the measurement noise matrix R be nonsingular. In other words, each of the measurements y_i must be corrupted by at least *some* noise for this method to work.

■ EXAMPLE 3.2

We return to our original problem of trying to estimate the resistance x of an unmarked resistor on the basis of k noisy measurements from a multimeter. In this case, x is a scalar so our k noisy measurements are given as

$$\begin{aligned}
y_i &= x + v_i \\
E(v_i^2) &= \sigma_i^2 \quad (i = 1, \ldots, k)
\end{aligned} \tag{3.16}$$

The k measurement equation can be combined into a single matrix equation as

$$\begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} x + \begin{bmatrix} v_1 \\ \vdots \\ v_k \end{bmatrix} \tag{3.17}$$

and the measurement noise covariance is given as

$$R = \text{diag}(\sigma_1^2, \ldots, \sigma_k^2) \tag{3.18}$$

Equation (3.15) shows that the optimal estimate of the resistance x is given as

$$\begin{aligned}
\hat{x} &= (H^T R^{-1} H)^{-1} H^T R^{-1} y \\
&= \left(\begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_k^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \times \\
& \quad \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_k^2 \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} \\
&= \left(\sum 1/\sigma_i^2 \right)^{-1} \left(y_1/\sigma_1^2 + \cdots + y_k/\sigma_k^2 \right)
\end{aligned} \tag{3.19}$$

We see that the optimal estimate $\hat{x}$ is a weighted sum of the measurements, where each measurement is weighted by the inverse of its uncertainty. In other words, we put more emphasis on certain measurements, in agreement